

NT-C-025 | Generative AI | Course Knowledge Profile

Identity, purpose, prerequisites, scope and boundaries

Category	Data & AI
Target learners	Developers, analysts and professionals building evidence-based AI-assisted workflows.
Prerequisites	Basic Python or JavaScript and API concepts are helpful.
Approved duration	10-14 weeks / 100-160 guided hours.
Final outcome	Build and evaluate a safe generative-AI application using prompts, retrieval, structured outputs and documented controls.
Assessment profile	Data/AI

Approved tools / platforms

Python or JavaScript, LLM API, embeddings/vector store, evaluation tools, Git. Provider and model must be version-recorded.

Knowledge boundaries

Include	Quality rule	Exclude / escalate
The eight approved modules, four sections, named tools, projects and mapped entry-level roles.	Require data provenance, cleaning decisions, train/test separation, suitable metrics, reproducibility and responsible interpretation.	Reject data leakage, invented metrics, unsupported causal claims and hidden manual corrections.
Current product/platform procedures only when version and date are known.	Use primary documentation and record the source version.	Do not invent current commands, prices, tax rates, policies or certification status.
Evidence-based career guidance.	Cite tests, projects, capstone, viva and verified resume evidence.	No guaranteed job, placement, salary, interview or employer claim.

LLM answer pattern: identify the course and version, answer from approved records, cite record IDs, distinguish verified facts from guidance, and state any missing evidence.

NT-C-025 | Generative AI | NT-C-025-S01 Deep Knowledge

Modules 1-2 | objectives, concepts, evidence and remediation

Section competency	Models, tokens, context, limitations and use cases; Instructions, examples, constraints and structured outputs
Completion evidence	Use-case analysis; Prompt test set
Section weight	20% of the weighted mock-test average
Assessment	10 concept MCQ (30 marks) + 4 interpretation scenarios (30) + 1 data/model practical (40)

Module 1 - GenAI foundation

Objective: Models, tokens, context, limitations and use cases

Observable evidence: Use-case analysis

Concept	Approved knowledge statement	Application behaviour	Evidence
Models	A defined competency within Generative AI. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when models is appropriate; apply it in a Generative AI task; verify the result.	Use-case analysis
tokens	A defined competency within Generative AI. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when tokens is appropriate; apply it in a Generative AI task; verify the result.	Use-case analysis
context	A defined competency within Generative AI. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when context is appropriate; apply it in a Generative AI task; verify the result.	Use-case analysis
limitations	A defined competency within Generative AI. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when limitations is appropriate; apply it in a Generative AI task; verify the result.	Use-case analysis
use cases	A defined competency within Generative AI. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when use cases is appropriate; apply it in a Generative AI task; verify the result.	Use-case analysis

Module 2 - Prompt design

Objective: Instructions, examples, constraints and structured outputs

Observable evidence: Prompt test set

Concept	Approved knowledge statement	Application behaviour	Evidence
Instructions	A defined competency within Generative AI. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when instructions is appropriate; apply it in a Generative AI task; verify the result.	Prompt test set
examples	A defined competency within Generative AI. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when examples is appropriate; apply it in a Generative AI task; verify the result.	Prompt test set
constraints	A defined competency within Generative AI. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when constraints is appropriate; apply it in a Generative AI task; verify the result.	Prompt test set
structured outputs	A defined competency within Generative AI. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when structured outputs is appropriate; apply it in a Generative AI task; verify the result.	Prompt test set

Remediation: If the learner cannot explain the purpose, complete the stated task or provide Use-case analysis; Prompt test set, return to Modules 1-2, complete one guided practice and then use a new equivalent test form.

NT-C-025 | Generative AI | NT-C-025-S02 Deep Knowledge

Modules 3-4 | objectives, concepts, evidence and remediation

Section competency	Requests, authentication, errors and logging; Documents, chunking, embeddings and vector search
Completion evidence	API application; RAG prototype
Section weight	25% of the weighted mock-test average
Assessment	10 concept MCQ (30 marks) + 4 interpretation scenarios (30) + 1 data/model practical (40)

Module 3 - API integration

Objective: Requests, authentication, errors and logging

Observable evidence: API application

Concept	Approved knowledge statement	Application behaviour	Evidence
Requests	A defined competency within Generative AI. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when requests is appropriate; apply it in a Generative AI task; verify the result.	API application
authentication	Verification of identity. In Generative AI, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when authentication is appropriate; apply it in a Generative AI task; verify the result.	API application
errors	A defined competency within Generative AI. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when errors is appropriate; apply it in a Generative AI task; verify the result.	API application
logging	A defined competency within Generative AI. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when logging is appropriate; apply it in a Generative AI task; verify the result.	API application

Module 4 - Retrieval

Objective: Documents, chunking, embeddings and vector search

Observable evidence: RAG prototype

Concept	Approved knowledge statement	Application behaviour	Evidence
Documents	A defined competency within Generative AI. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when documents is appropriate; apply it in a Generative AI task; verify the result.	RAG prototype
chunking	A defined competency within Generative AI. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when chunking is appropriate; apply it in a Generative AI task; verify the result.	RAG prototype
embeddings	A numerical representation used to compare semantic similarity. In Generative AI, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when embeddings is appropriate; apply it in a Generative AI task; verify the result.	RAG prototype
vector search	A defined competency within Generative AI. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when vector search is appropriate; apply it in a Generative AI task; verify the result.	RAG prototype

Remediation: If the learner cannot explain the purpose, complete the stated task or provide API application; RAG prototype, return to Modules 3-4, complete one guided practice and then use a new equivalent test form.

NT-C-025 | Generative AI | NT-C-025-S03 Deep Knowledge

Modules 5-6 | objectives, concepts, evidence and remediation

Section competency	Test sets, relevance, accuracy and human review; Prompt injection, data handling and guardrails
Completion evidence	Evaluation report; Risk-control checklist
Section weight	25% of the weighted mock-test average
Assessment	10 concept MCQ (30 marks) + 4 interpretation scenarios (30) + 1 data/model practical (40)

Module 5 - Evaluation

Objective: Test sets, relevance, accuracy and human review

Observable evidence: Evaluation report

Concept	Approved knowledge statement	Application behaviour	Evidence
Test sets	A defined competency within Generative AI. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when test sets is appropriate; apply it in a Generative AI task; verify the result.	Evaluation report
relevance	A defined competency within Generative AI. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when relevance is appropriate; apply it in a Generative AI task; verify the result.	Evaluation report
accuracy	A defined competency within Generative AI. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when accuracy is appropriate; apply it in a Generative AI task; verify the result.	Evaluation report
human review	A defined competency within Generative AI. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when human review is appropriate; apply it in a Generative AI task; verify the result.	Evaluation report

Module 6 - Safety and privacy

Objective: Prompt injection, data handling and guardrails

Observable evidence: Risk-control checklist

Concept	Approved knowledge statement	Application behaviour	Evidence
Prompt injection	The instructions and context supplied to a generative model. In Generative AI, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when prompt injection is appropriate; apply it in a Generative AI task; verify the result.	Risk-control checklist
data handling	A defined competency within Generative AI. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when data handling is appropriate; apply it in a Generative AI task; verify the result.	Risk-control checklist
guardrails	A defined competency within Generative AI. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when guardrails is appropriate; apply it in a Generative AI task; verify the result.	Risk-control checklist

Remediation: If the learner cannot explain the purpose, complete the stated task or provide Evaluation report; Risk-control checklist, return to Modules 5-6, complete one guided practice and then use a new equivalent test form.

NT-C-025 | Generative AI | NT-C-025-S04 Deep Knowledge

Modules 7-8 | objectives, concepts, evidence and remediation

Section competency	Tool use, agents and approval points; Monitoring, cost, documentation and responsible deployment
Completion evidence	Controlled workflow; Evaluated GenAI product
Section weight	30% of the weighted mock-test average
Assessment	10 concept MCQ (30 marks) + 4 interpretation scenarios (30) + 1 data/model practical (40)

Module 7 - Workflow automation

Objective: Tool use, agents and approval points

Observable evidence: Controlled workflow

Concept	Approved knowledge statement	Application behaviour	Evidence
Tool use	A defined competency within Generative AI. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when tool use is appropriate; apply it in a Generative AI task; verify the result.	Controlled workflow
agents	A defined competency within Generative AI. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when agents is appropriate; apply it in a Generative AI task; verify the result.	Controlled workflow
approval points	A defined competency within Generative AI. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when approval points is appropriate; apply it in a Generative AI task; verify the result.	Controlled workflow

Module 8 - Capstone

Objective: Monitoring, cost, documentation and responsible deployment

Observable evidence: Evaluated GenAI product

Concept	Approved knowledge statement	Application behaviour	Evidence
Monitoring	Collection and interpretation of metrics, logs and events to understand system health. In Generative AI, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when monitoring is appropriate; apply it in a Generative AI task; verify the result.	Evaluated GenAI product
cost	A defined competency within Generative AI. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when cost is appropriate; apply it in a Generative AI task; verify the result.	Evaluated GenAI product
documentation	A defined competency within Generative AI. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when documentation is appropriate; apply it in a Generative AI task; verify the result.	Evaluated GenAI product
responsible deployment	A defined competency within Generative AI. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when responsible deployment is appropriate; apply it in a Generative AI task; verify the result.	Evaluated GenAI product

Remediation: If the learner cannot explain the purpose, complete the stated task or provide Controlled workflow; Evaluated GenAI product, return to Modules 7-8, complete one guided practice and then use a new equivalent test form.

NT-C-025 | Generative AI | Skill Ontology & Dependency Map

Atomic skills, learning order and evidence strength

Skill ID	Competency	Depends on	Evidence	Type
NT-C-025-SK01	GenAI foundation	Prerequisite foundation	Use-case analysis	Critical
NT-C-025-SK02	Prompt design	M01 plus earlier foundations	Prompt test set	Critical
NT-C-025-SK03	API integration	M02 plus earlier foundations	API application	Supporting
NT-C-025-SK04	Retrieval	M03 plus earlier foundations	RAG prototype	Supporting
NT-C-025-SK05	Evaluation	M04 plus earlier foundations	Evaluation report	Supporting
NT-C-025-SK06	Safety and privacy	M05 plus earlier foundations	Risk-control checklist	Critical
NT-C-025-SK07	Workflow automation	M06 plus earlier foundations	Controlled workflow	Supporting
NT-C-025-SK08	Capstone	M07 plus earlier foundations	Evaluated GenAI product	Critical

Evidence-strength hierarchy

Strength	Evidence source	Use
High	Trainer-observed practical, viva, working source/project files and reproducible output.	May support verified mastery when rubric threshold is met.
Medium	Scored mock tests, reviewed assignments and documented portfolio artifacts.	Supports mastery with corroboration.
Low	Resume claim, self-report, attendance or video completion without demonstration.	May trigger verification; cannot alone support Apply Now.

CareerTours must not treat attendance, time spent or content opened as skill mastery.

NT-C-025 | Generative AI | Assessment Knowledge & Sample Items

Non-live examples for LLM training and evaluation

Sample ID	Type	Question	Approved answer / rubric anchor
NT-C-025-S01-EX01	Evidence/application	What evidence best demonstrates the combined competency of GenAI foundation and Prompt design?	The learner should provide Use-case analysis; Prompt test set and explain how the work applies Models, tokens, context, limitations and use cases; Instructions, examples, constraints and structured outputs. Recall alone is insufficient.
NT-C-025-S01-EX02	Scenario/remediation	A learner reports completing Prompt design but cannot produce Prompt test set. What should CareerTours do?	Mark the evidence Not Yet Verified, assign practice from NT-C-025-S01, request Prompt test set, and allow a new equivalent test only after remediation.
NT-C-025-S02-EX01	Evidence/application	What evidence best demonstrates the combined competency of API integration and Retrieval?	The learner should provide API application; RAG prototype and explain how the work applies Requests, authentication, errors and logging; Documents, chunking, embeddings and vector search. Recall alone is insufficient.
NT-C-025-S02-EX02	Scenario/remediation	A learner reports completing Retrieval but cannot produce RAG prototype. What should CareerTours do?	Mark the evidence Not Yet Verified, assign practice from NT-C-025-S02, request RAG prototype, and allow a new equivalent test only after remediation.
NT-C-025-S03-EX01	Evidence/application	What evidence best demonstrates the combined competency of Evaluation and Safety and privacy?	The learner should provide Evaluation report; Risk-control checklist and explain how the work applies Test sets, relevance, accuracy and human review; Prompt injection, data handling and guardrails. Recall alone is insufficient.
NT-C-025-S03-EX02	Scenario/remediation	A learner reports completing Safety and privacy but cannot produce Risk-control checklist. What should CareerTours do?	Mark the evidence Not Yet Verified, assign practice from NT-C-025-S03, request Risk-control checklist, and allow a new equivalent test only after remediation.
NT-C-025-S04-EX01	Evidence/application	What evidence best demonstrates the combined competency of Workflow automation and Capstone?	The learner should provide Controlled workflow; Evaluated GenAI product and explain how the work applies Tool use, agents and approval points; Monitoring, cost, documentation and responsible deployment. Recall alone is insufficient.
NT-C-025-S04-EX02	Scenario/remediation	A learner reports completing Capstone but cannot produce Evaluated GenAI product. What should CareerTours do?	Mark the evidence Not Yet Verified, assign practice from NT-C-025-S04, request Evaluated GenAI product, and allow a new equivalent test only after remediation.

Live bank rule: Generate at least three times the live quantity for each section using the approved Data/AI blueprint.
Human-review every scored item before initial production release.

NT-C-025 | Generative AI | Labs, Projects & Scoring Rubrics

Practical proof of learning and ownership

Level	Approved project	Skills evidenced	Required deliverables
Mini 1	Structured assistant	Prompting and JSON output	Brief; working/source file or reproducible environment; output; validation notes; reflection.
Mini 2	Document Q&A;	Retrieval and citations	Brief; working/source file or reproducible environment; output; validation notes; reflection.
Capstone	Career-support prototype	Evaluation, safety and monitoring	Brief; working/source file or reproducible environment; output; validation notes; reflection.

Standard project rubric

Dimension	Weight	What earns credit
Problem framing	15%	Clear objective, user/business need, assumptions and success criteria.
Technical/design accuracy	25%	Correct methods, appropriate tools and sound decisions.
Completeness	15%	Required features, assets, data or workflow are present.
Testing and validation	15%	Important normal, edge, quality or failure cases are checked.
Evidence and reproducibility	15%	Source/working files, setup, inputs, outputs and version information are available.
Documentation	5%	Concise structure, instructions and limitations.
Viva / ownership	10%	Learner can explain decisions, demonstrate work and respond to a change request.

Require data provenance, cleaning decisions, train/test separation, suitable metrics, reproducibility and responsible interpretation.

NT-C-025 | Generative AI | Career Roles, Evidence & Gap Actions

Explainable top-three occupation matching

Rank	Occupation	Default weighted skills	Minimum evidence	Gap action
1	Generative AI Application Trainee	API 34% [critical]; prompting 33% [critical]; evaluation 33%	Working prototype	If a critical skill is below 60 or working prototype is missing, return Improve First or Needs Evidence with the linked module/project.
2	AI Automation Associate	Workflows 34% [critical]; tools 33% [critical]; controls 33%	Automated process demo	If a critical skill is below 60 or automated process demo is missing, return Improve First or Needs Evidence with the linked module/project.
3	LLM Operations Trainee	Testing 34% [critical]; monitoring 33% [critical]; documentation 33%	Evaluation report	If a critical skill is below 60 or evaluation report is missing, return Improve First or Needs Evidence with the linked module/project.

Course readiness	45% section mocks + 20% mini projects + 25% capstone + 10% viva.
Role readiness	45% role-relevant mastery + 25% projects/capstone + 20% aligned mocks + 10% verified resume evidence.
Evidence confidence	Show separately; Apply Now requires at least 70.
Resume extraction	Use only relevant skills, projects, education, experience and portfolio evidence; minimise personal data before external model processing.

A student may be Apply Now for one mapped occupation and Improve First for another. Return independent evidence and gaps for each role.

NT-C-025 | Generative AI | FAQs, Misconceptions & LLM Response Pack

Course-specific retrieval and response patterns

Approved FAQ	Answer
Who is Generative AI for?	Developers, analysts and professionals building evidence-based AI-assisted workflows.
What should I know before starting?	Basic Python or JavaScript and API concepts are helpful.
How long is the approved learning journey?	10-14 weeks / 100-160 guided hours.
What will I be able to demonstrate?	Build and evaluate a safe generative-AI application using prompts, retrieval, structured outputs and documented controls.
Does completion guarantee a job?	No. Completion and role readiness are evidence-based guidance; employment, placement, salary and employer selection are not guaranteed.
What if I fail a section?	CareerTours identifies the failed skill tags and module, assigns targeted practice, and permits an equivalent retest after remediation.

Misconception	Approved correction
Attendance proves mastery	False - require tests and observable practical/project evidence.
One total score proves every role fit	False - calculate each occupation using its weighted skills and evidence.
The LLM can add missing details	False - retrieve approved records or state that evidence is insufficient.
Course completion guarantees employment	False - never claim guaranteed job, placement, salary or interview.
Generative AI knowledge never changes	False - version-sensitive tools, laws, taxes, security and platform procedures require review.

Retrieval metadata

```
course_code=NT-C-025; corpus_version=3.0;
record_types=course_profile,module,section,skill,assessment_sample,project,occupation_mapping,faq,misconception;
approval_status=approved; effective_date=2026-08-12
```